AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 12 · WORKBOOK

Debug the Cause

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected debugging record.

Locate a mismatch, distinguish hypotheses, reject a patch, try the ticket case, then record repair evidence.

Keep one habit: Keep expected and observed values in separate columns.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Repeating Efficiently. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Algorithms as Recipes. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Sound Becomes a Signal. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Repeating Efficiently

Algorithms as Recipes

Sound Becomes a Signal

W1 · Locate the first divergence

> Calculate the intended trace before judging the later symptom.

Your evidence record

- A procedure must add 2, double the updated value, then subtract 1. Starting at 3, an observed **trace** is 3, 5, 8, 7.

  Write the **expected trace**, identify the first difference and explain why the final subtraction is not necessarily a separate fault.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Choose a revealing new input

> Predict each suspected fault before running a comparison.

Your evidence record

- In W1 the faulty middle step may add 3, or add the original starting value. Both explain 5 becoming 8 when the start is 3. Choose start 1 or start 3 for a revealing new test.

  Predict the final result under each explanation and under the intended procedure.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Reject a constant-output patch

> Give a new input with a different specified result.

Your evidence record

- A learner repairs the original points machine by reporting 12 for every **input**, because 12 is correct from start 4. Give a different starting **input** and its **expected output** under add 3, double, subtract 2.

  Explain why the constant report is not a valid repair and what earlier **evidence** should be preserved.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Debug a new ticket rule

> Keep one first attempt before hints or feedback.

Read and explain independently

A paper ticket rule must add 1 to the **input** count. If the updated count is at least 5, report Group; otherwise report Single. A copied version says above 5.

Choose the smallest three consecutive nonnegative starting counts that test just below, exactly at and just above the comparison boundary. Predict both versions, identify the revealing start and state the repair and retests.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Debug a new ticket rule

W5 · Report after the first W4 attempt and feedback

> Record the exact repair, retests and limits after trying the reserved ticket case.

Your evidence record

- After repairing W4, all three chosen tests pass. Write a short report with the fault, the exact repair, the three **input**s and **expected output**s, and what those passing tests establish.

  Explain why saying every possible error is now impossible would go beyond the **evidence**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

When the sorter fails, log the first wrong stage and change only the responsible rule, **input**, or label.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Choose between two explanations for Debugging with revealing tests. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Debugging with revealing tests. Proposed explanation: “The last wrong number locates the fault”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Starting at square 0, the target is square 3. Program: repeat twice {MOVE 1}; STOP. A teammate changes the target to 2 and calls the bug fixed. Repair the program and give two tests.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

A robot should go from square one to square three. Its program is MOVE ONE, STOP.

Find the cause and repair one thing.

Show your **trace**, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A sorting rule says blue goes left. A blue card goes right. The reader correctly reports blue, but the direction rule says blue goes right. Which part should you fix first?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 12 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the **expected** answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

The requirement is to move exactly three squares forward from any valid starting square, then stop. Starting at square 0, repeat twice {MOVE 1}; STOP ends at 2. A teammate reduces the required distance to two squares.

Repair the program without changing the requirement; test starts 0 and 2.

> **My explanation after practice**

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

A paper ticket rule must add 1 to the **input** count. If the updated count is at least 5, report Group; otherwise report Single. A copied version says above 5.

Choose the smallest three consecutive nonnegative starting counts that test just below, exactly at and just above the comparison boundary. Predict both versions, identify the revealing start and state the repair and retests.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

> Keep the explanation and the support you used.

Use this when helpful

A fictional paper calculator should double the current value, but a copied instruction behaves strangely.

After **input** 6, the faulty step reports 12. Hypothesis A: add 6. Hypothesis B: double.

Both hypotheses fit this first **observation**. A new **input** may be chosen.

AI ACADEMY / AI-1/12

A revealing test separates explanations

	Input	Add 6 predicts	Double predicts
Original input	6	12	12
Recorded new input	2	8	4
Proposed input	0	6	0

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/12

Debug the Cause

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Observe

Input 6 produced 12; that fact fits two explanations.



2. Predict alternatives

For input 2, adding 6 predicts 8; doubling predicts 4.



3. Run revealing test

The fictional new record is input 2 -> output 8.



4. Locate cause

This record supports add-6 over doubling for this comparison.

CHANGE ONE THING

Use input 6 again instead of 2.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Propose **input** 0 and give both predicted **output**s without inventing an **observed result**.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: When the sorter fails, write competing explanations and predicted **trace**s before changing code or instructions.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Make a transparent first version. Keep a **trace** of how one **input** becomes an **output**; explain the rule or learned component.

When the sorter fails, write competing explanations and predicted **trace**s before changing code or instructions.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional paper calculator should double the current value, but a copied instruction behaves strangely.
- Which labelled step matches this detail? This record supports add-6 over doubling for this comparison.
- 1. Observe
- 2. Run revealing test
- 3. Predict alternatives
- 4. Locate cause

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Locate cause This record supports add-6 over doubling for this comparison. Repeating **input** 6 would not separate the two hypotheses because both again predict 12.

Input 2 makes their **prediction**s different before the test is run. The second record supports investigating an added constant, but inspecting the instruction is still needed to confirm the implementation.

Changed-case question

- Changed-case practice: Use **input** 6 again instead of 2. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. Both hypotheses predict 12; the test remains inconclusive.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial **trace**. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Both hypotheses predict 12; the test remains inconclusive. A revealing test requires different predicted **output**s, not merely another recorded run.

Repair and save your work

Review Repeating Efficiently (ai-1/11). Goal: **Trace** a loop by testing its condition before every repetition. Short practice: **Trace** a fresh start of 9. How many additions occur, and must the report equal 10?.

Return to this week and explain what you changed.

Review Algorithms as Recipes (ai-1/9). Goal: **Trace** an ordered algorithm using the current value after every instruction. Short practice: Use the arrows to explain where the two zero-start runs first differ..

Return to this week and explain what you changed.

Review Sound Becomes a Signal (ai-1/6). Goal: Interpret an ordered sequence of sound-signal samples with a stated time interval. Short practice: Does the all-zero retained record prove the full signal was silent?

Use the picture.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Debug the Cause: the case

AI ACADEMY / AI-1/12 · 01 · LOOK AT THE FACTS

Debug the Cause: the case

The situation

A fictional paper calculator should double the current value, but a copied instruction behaves strangely.

Supplied evidence

After input 6, the faulty step reports 12. Hypothesis A: add 6. Hypothesis B: double. Both hypotheses fit this first observation. A new input may be chosen.

By the end, I can choose a test that separates two possible causes of an error.

Follow the labelled picture with the matching explanation. Open the vector diagram

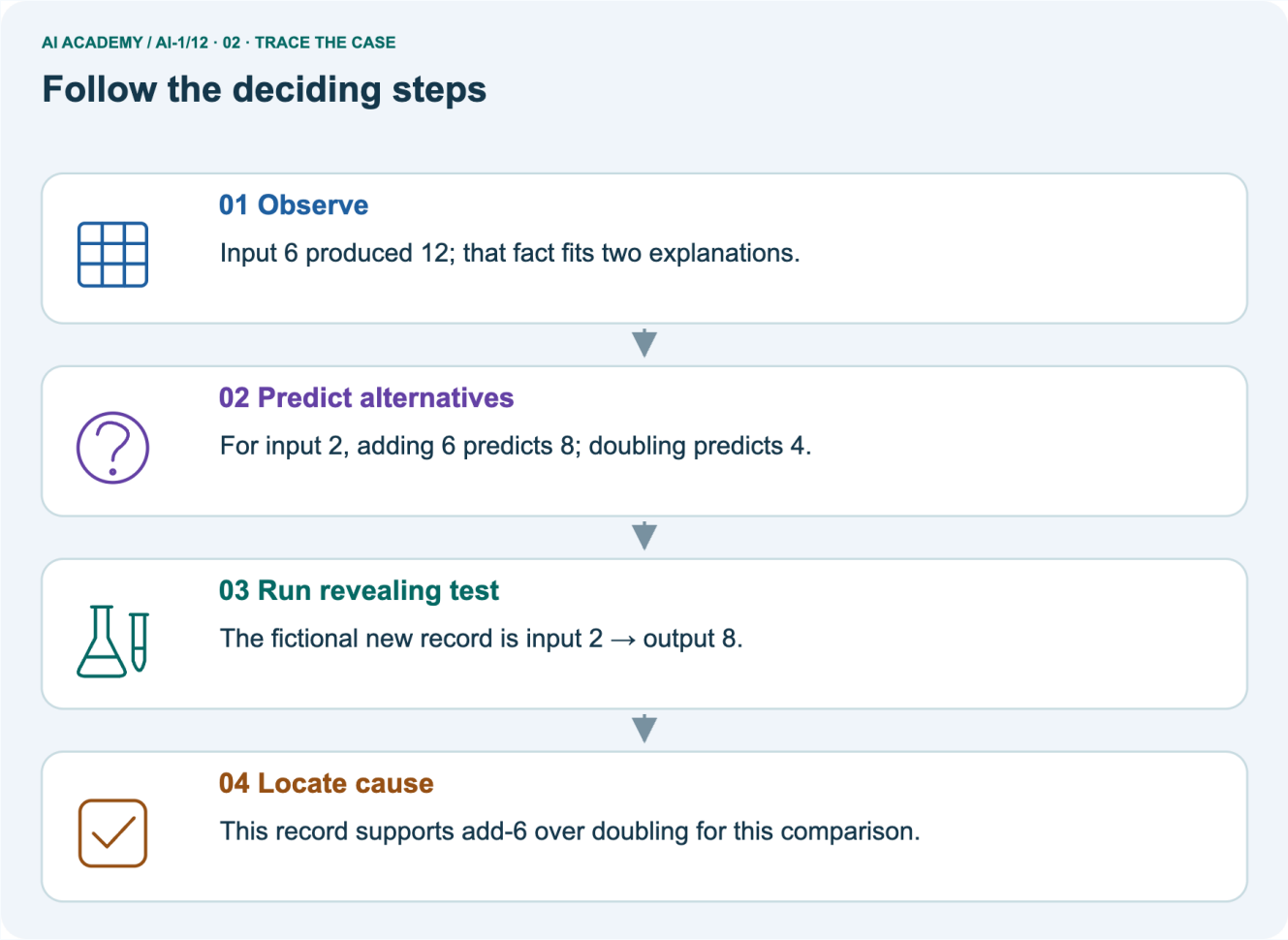
A fictional paper calculator should double the current value, but a copied instruction behaves strangely. Supplied facts: After **input** 6, the faulty step reports 12. Hypothesis A: add 6. Hypothesis B: double.

Both hypotheses fit this first **observation**. A new **input** may be chosen.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



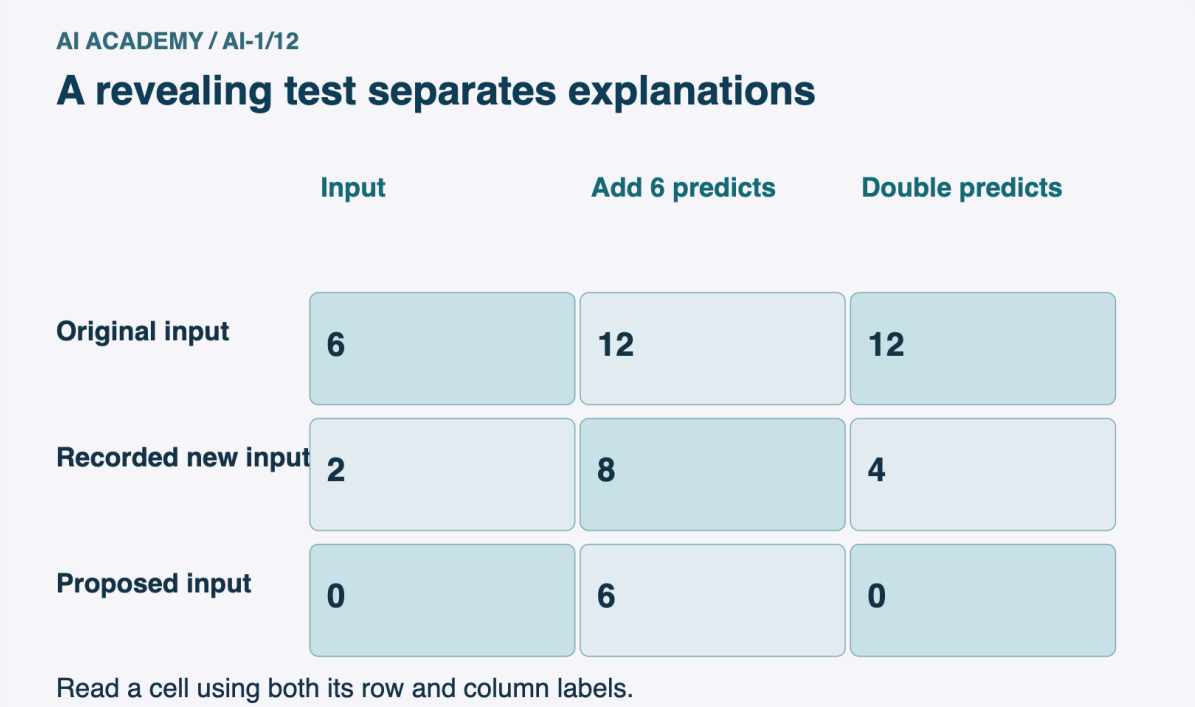
Follow the labelled picture with the matching explanation. Open the vector diagram

Observe: **Input** 6 produced 12; that fact fits two explanations. → Predict alternatives: For **input** 2, adding 6 predicts 8; doubling predicts 4. → Run revealing test: The fictional new record is **input** 2 → **output** 8. → Locate cause: This record supports add-6 over doubling for this comparison.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. A revealing test separates explanations



Follow the labelled picture with the matching explanation. Open the vector diagram

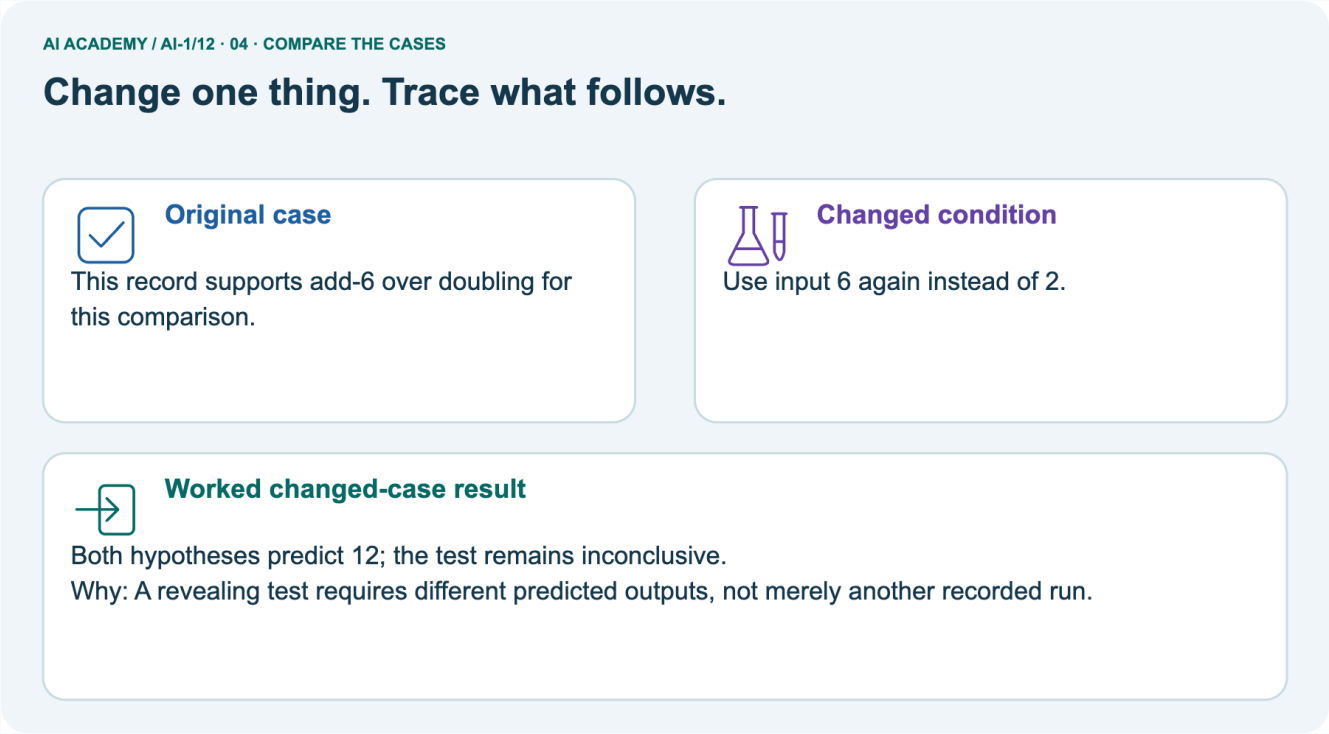
A revealing test separates explanations. Original **input**: **Input** = 6, Add 6 predicts = 12, Double predicts = 12. Recorded new **input**: **Input** = 2, Add 6 predicts = 8, Double predicts = 4.

Proposed **input**: **Input** = 0, Add 6 predicts = 6, Double predicts = 0.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

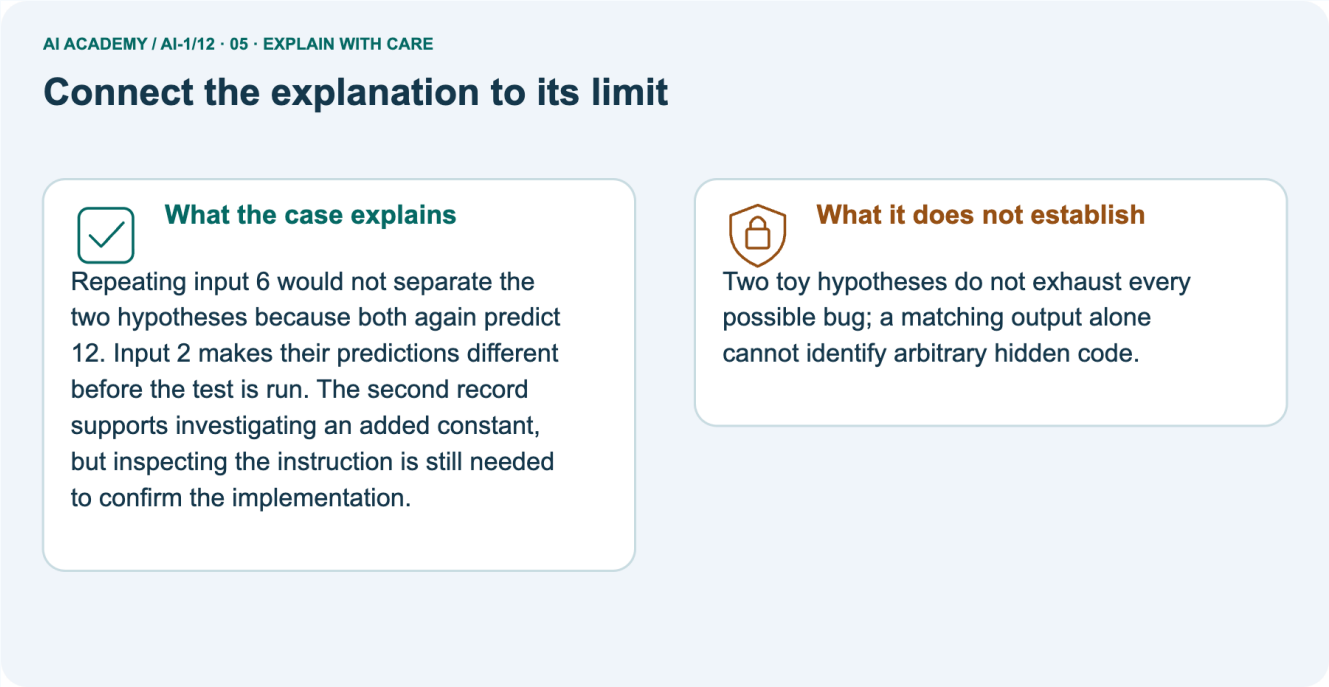
Original case: This record supports add-6 over doubling for this comparison. Changed condition: Use **input** 6 again instead of 2. **Expected result**: Both hypotheses predict 12; the test remains inconclusive.

Reason: A revealing test requires different predicted **output**s, not merely another recorded run.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Repeating **input** 6 would not separate the two hypotheses because both again predict 12. **Input** 2 makes their **prediction**s different before the test is run.

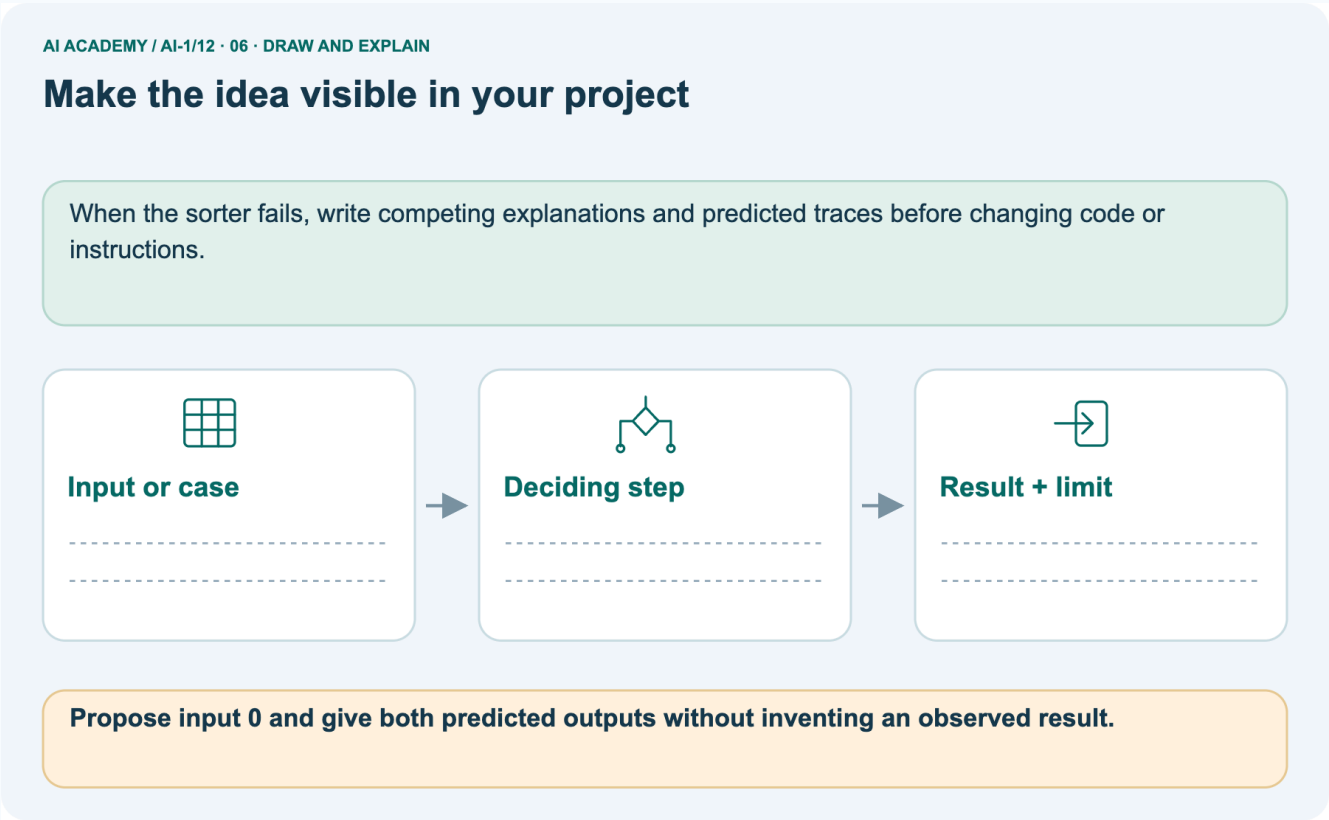
The second record supports investigating an added constant, but inspecting the instruction is still needed to confirm the implementation.

Limit: Two toy hypotheses do not exhaust every possible bug; a matching **output** alone cannot identify arbitrary hidden code.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

When the sorter fails, write competing explanations and predicted **trace**s before changing code or instructions. Draw your **input**, deciding step, result and limit.

Supported question: Propose **input** 0 and give both predicted **output**s without inventing an **observed result**.

Draw your project **trace** and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Debug the Cause: the case

02. Follow the deciding steps

03. A revealing test separates explanations

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.